

SmartCity AR: Re-imagining Cape Town's Foreshore Freeway

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ABSTRACT

Urban planning requires a comprehensive understanding of infrastructure, transportation, and sustainability. Traditional tools like static maps lack interactivity, limiting effective decision-making. This report introduces SmartCity AR, an Augmented Reality-based urban planning viewer designed to enhance visualization and analysis. The application, which focuses on Cape Town's Foreshore Freeway Precinct, targets traffic congestion and wasteful land use by allowing users to compare sustainable and unsustainable development scenarios, explore transportation hubs, and map pedestrian-friendly pathways. The paper covers the system's design, user requirements, methodology, and implementation techniques.

1 INTRODUCTION

The Foreshore Freeway Precinct, centrally located within Cape Town, exemplifies the urban challenges brought on by rapid population growth and increasing traffic demands. Acting as a key link between the central business district and nearby areas, the precinct struggles with daily traffic jams, poor use of land, and limited infrastructure for pedestrians. These challenges point to the need for fresh, human-focused development ideas. However, conventional planning tools—like maps and static visuals—often fail to clearly present these plans to both officials and the public [1].

Cape Town's population has surpassed 5 million as of 2025, with rapid urbanization and a 1.72% annual growth rate [1]. This growth has intensified pressure on key inner-city areas like the Foreshore Freeway Precinct — a zone already plagued by chronic congestion, underutilized land, and limited pedestrian accessibility. While city-wide traffic congestion costs drivers an average of 94 hours per year, the impact is particularly visible in this central corridor, which serves as a crucial link between the CBD and surrounding districts [2].

To address this, the project proposes an Augmented Reality (AR)-based educational tool that enables urban planners, architects, and stakeholders to visualize and interact with different infrastructure designs in an immersive environment. By overlaying proposed enhancements onto the real-world precinct, users can better understand how strategic planning decisions—such as improved public transport integration, MyCiTi bus expansion, and optimized land use—can enhance connectivity and sustainability.

2 LITERATURE REVIEW

Augmented Reality (AR) has emerged as a powerful tool for urban planning, allowing users to overlay digital models onto real-world environments. Previous studies have demonstrated the potential of AR in infrastructure design, transportation planning, and public engagement. CityBuildAR's mobile app demonstrated that AR could engage the public in placemaking, enabling users

with limited design knowledge to actively contribute to urban planning and express preferences, such as favouring more green spaces in public areas [3]. AR's effectiveness in participatory urban planning has also been evaluated in studies that show varying preferences for levels of detail in 3D models, with higher detail offering more clarity but introducing challenges like visual clutter and scaling issues [4]. In architecture and urban planning, AR has been used to visualize both simple and complex design elements, with techniques like GPS-based and marker-based AR offering dynamic ways to enhance user interaction and spatial understanding in both professional and public settings [5]. While these studies highlight AR's effectiveness, few implementations specifically target sustainable urban planning.

This project hypothesizes that such an AR tool will improve user comprehension and support better-informed planning choices compared to traditional 2D methods. Platforms like Unity are widely used for AR development due to their cross-platform support and ease of integration with AR Foundation but may require optimization for performance on mobile devices. While VR headsets provide immersive experiences ideal for simulation, they are less practical for public engagement due to cost and accessibility. AR headsets like HoloLens offer on-site visualization benefits but are expensive and have limited battery life. Given these trade-offs, Unity with mobile AR offers a practical balance for accessible, scalable urban planning tools.

3 PROJECT PROPOSAL

Urban planning requires a clear understanding of how infrastructure and transport networks interact within a city. Traditional tools—such as static maps (e.g. paper maps or 2D blueprints) and static diagrams (e.g., fixed architectural renderings or zoning diagrams)—often fail to provide an intuitive and interactive way to visualize these complex relationships. Static maps are limited in their ability to demonstrate dynamic features like traffic flow, pedestrian movement, and real-time infrastructure changes, while static diagrams can be overly technical and difficult for non-experts to interpret. This makes it challenging for urban planners, architects, and the public to assess potential improvements in city infrastructure and urban development effectively.

In Cape Town's Foreshore Freeway Precinct, traffic congestion, limited pedestrian access, and inefficient transport integration remain significant challenges [6]. Additionally, the precinct's potential for mixed-use developments and underutilized spaces beneath the freeway has not been fully explored. The precinct's unfinished infrastructure, like the incomplete highway in Figure C (appendix C), causes traffic congestion and poor land use. Redevelopment plans in Figures D and E (Appendix D), including the CITYLIFT green city district and affordable housing initiatives, guide the AR tool's design by offering real planning scenarios to simulate. While various redevelopment ideas have

been proposed, there is a clear need for an accessible visualization tool that demonstrates how different urban design choices—such as public transport expansion, improved MyCiTi bus integration, pedestrian-friendly infrastructure, and optimized road networks—can improve mobility, connectivity, and the overall urban environment. This project will develop an AR-based educational tool that allows users to do the following.

- Visualize proposed enhancements to the Foreshore Freeway Precinct, such as improved pedestrian walkways, expanded public transport links, and mixed-use developments beneath the freeway. This helps stakeholders see the impact of spatial redesigns in real-world context.
- Toggle infrastructure layers to explore how various urban elements—like roads, transport hubs, public spaces, and buildings—interact with each other. This feature promotes a deeper understanding of spatial relationships and planning complexity.
- Compare sustainable vs. unsustainable planning approaches, showcasing how strategic design interventions can reduce congestion, enhance connectivity, and promote long-term urban sustainability.

By offering localized, interactive visualizations, the tool aims to help urban planners, students, and the public better understand how proposed changes such as MyCiTi bus integration and pedestrian-friendly infrastructure can improve mobility, accessibility, and the liveability of the Foreshore Freeway Precinct.

4 USER REQUIREMENTS

4.1 About the Project

4.1.1 Purpose. Urban planning is a complex process that requires effective visualization and simulation of different city infrastructure scenario [7]. Traditionally, planners have relied on static maps and blueprints to represent their ideas, but these tools do not provide the interactivity needed to truly understand the dynamic nature of urban systems. This project aims to develop an Augmented Reality based educational tool that helps urban planners, architects, and civil engineers visualize city layouts and proposed infrastructure changes in real-time. The tool will specifically focus on the Foreshore Freeway Precinct in Cape Town, which suffers from issues like traffic congestion, inefficient land use, and poor pedestrian access. The goal is to provide an immersive AR experience where users can visualize and interact with sustainable and unsustainable planning scenarios, facilitating better decision-making in urban development.

4.1.2 Agile Development Approach. This project follows an Agile iterative development cycle, with weekly milestones to incrementally test and refine core features [8].

4.2 User View of the Project

The primary users of this project are urban planners, architects, civil engineers, and students of urban studies. These users need a tool that allows them to visualize proposed changes to city infrastructure interactively. SmartCity AR will allow them to see how different planning scenarios—such as better pedestrian pathways, public transport integration, and optimized land use—will affect the urban environment. The user can manipulate the city layout, toggle between data layers, and explore the impact of various sustainable versus unsustainable planning strategies. This tool will aid them in creating more efficient, connected, and sustainable urban spaces by providing an accurate representation of how changes in infrastructure will influence traffic flow, pedestrian movement, and overall city functionality.

4.3 Problem Domain

The problem domain of this project involves urban planning, city infrastructure, and the visualization of proposed developments in an easily interpretable manner. Urban planners often work with complex datasets, static models, and maps to make informed decisions, but these tools fail to effectively show the real-world implications of their planning decisions [7]. To address this gap, the project utilizes AR to bring the planning process into an interactive, visual, and immersive space. Users must be familiar with urban planning concepts such as land use, transportation networks, pedestrian infrastructure, and sustainability metrics. Additionally, the tool needs to integrate with 3D models, data layers (such as traffic flow), and real-time interactions to enhance the user experience.

4.4 Expectations

SmartCity AR is expected to serve as a comprehensive tool that enables urban planners and other users.

4.4.1 Visualize Proposed Developments. The tool should allow users to interact with and visualize city layouts in Cape Town, proposed changes, and urban designs in a 3D AR environment.

4.4.2 Scenario Simulation. The tool must allow users to simulate various urban planning scenarios (e.g. adding new transport hubs, modifying pedestrian routes, and redesigning public spaces in Cape Town).

4.4.3 Sustainability Comparison. Users should be able to compare the effects of sustainable and unsustainable planning approaches in terms of traffic flow, pedestrian access, and environmental impact in Cape Town [7].

4.4.4 Interactive Features. The viewer should support intuitive user interactions, such as zooming, rotating, and toggling between different data layers (e.g. traffic data, pedestrian data).

4.4.5 Ease of Use. The tool should provide a simple and accessible interface for users, prioritizing usability and functionality. The UI should be easy for both technical and non-technical users to navigate.

4.4.6 Educational Value. The tool should serve as an educational tool, helping users understand urban planning concepts, the impacts of different planning decisions, and how to design cities more sustainably and efficiently.

The primary objective of the SmartCity AR is to help urban planners make informed, data-driven decisions that lead to more sustainable, efficient, and well-connected urban spaces in Cape Town. It should enhance the planning process by providing a deeper understanding of how infrastructure interacts within the urban environment, making it easier to identify potential problems and solutions before they are implemented in the real world.

4.5 Out of Scope

4.5.1 iOS Support. The app will not support iOS devices in the current version. The focus will be on Android devices, and features such as iOS-specific AR capabilities (ARKit) are excluded.

4.5.2 Real-time Data Integration. The initial release will not include real-time traffic or environmental data, although this feature may be considered in future iterations of the application.

4.5.3 Advanced AI Simulation. The app will not include advanced artificial intelligence features for traffic or pedestrian movement prediction. The current focus will be on basic simulations and visualizations.

4.5.4 Multiplayer/Collaborative Features. The application will not feature multiplayer or collaborative capabilities in the current version.

4.5.5 Geographic Scope Limitation. The tool is specifically designed for the Foreshore Freeway Precinct in Cape Town. While the core system could be adapted for other cities, such expansion is outside the scope of the current version.

4.6 Use Case Diagram

The use case diagram in Figure 1 for SmartCity AR outlines the key interactions between the system and its users.

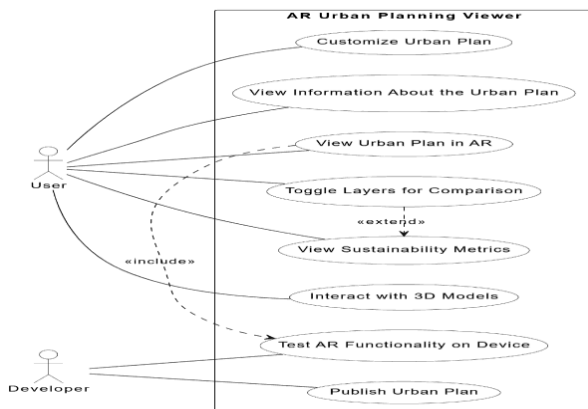


Figure 1: Use case diagram for SmartCity AR showcasing user interactions, system functionalities, and relationships between key actions.

5 REQUIREMENTS ANALYSIS

5.1 Current System

SmartCity AR is being developed on a Windows 11 platform using the Unity game engine for creating the 3D AR models, and AR Foundation for cross-platform AR support. The tool will also utilize Visual Studio for development, and GitHub for version control and collaboration. The system requires Android Studio for testing AR functionality on mobile devices. For users, the tool can be run on devices such as Android smartphones supporting ARCore. The development environment also includes the following prerequisites.

5.1.1 Unity 6.1. The latest version of the Unity engine is used to build and render the AR environment.

5.1.2 AR Foundation 5.0. This framework enables cross-platform AR development, supporting both Android and iOS.

5.1.3 Visual Studio 2022 with Unity integration. This integrated development environment provides tools for coding, debugging, and managing the project.

5.1.4 Git 2.7.4. A version control system is used to manage source code, track changes, and facilitate team collaboration.

5.1.5 Android Studio. This IDE is essential for testing AR functionality on mobile devices, ensuring compatibility with Android-based ARCore.

5.1.6 Blender. A 3D modelling software is used to create and optimize assets for the AR environment.

5.1.7 Itch.io. Once the development is complete, the tool will be published as an APK for Android and shared on platforms such as Itch.io.

All tools used in this project are free and open-source or available through free academic licenses, making the project cost-effective and accessible. Testing is conducted using Android devices that support ARCore, with on-campus access to mobile testing equipment where needed.

5.2 Functional Requirements

5.2.1 Basic City Layout Visualization. The app must allow the user to view a predefined 3D city model of Cape Town, which will include streets, buildings, roads, and other relevant urban infrastructure. Users should be able to interact with this city layout to simulate and compare different planning scenarios.

5.2.2 Building Manipulation. Users should be able to move, add, or remove buildings in the AR environment, allowing for flexible city planning and experimentation with urban design.

5.2.3 User Interactivity. The app will include intuitive touch-based interactions for navigation, such as zooming in/out, rotating

the city layout, and switching between different views of the model.

5.2.4 Scenario Simulation. The AR viewer will enable users to visualize various urban planning scenarios (e.g. pedestrian-friendly spaces, public transport integration, green corridors) and analyse their impact on traffic flow and accessibility.

5.2.5 Data Layer Toggle. The application will support toggling between different data layers, such as traffic data, pedestrian movement, and infrastructure utilization, to help users better understand the dynamics of urban spaces of Cape Town.

5.2.6 Sustainable vs. Unsustainable Comparison. The app must allow users to toggle between sustainable and unsustainable urban planning scenarios to analyse the effects of different planning strategies on the environment and public mobility in Cape Town.

5.3 Non-Functional Requirements

The non-functional requirements for the Urban Planning AR Viewer are as follows.

5.3.1 Performance. The application must load the main AR scene within 5 seconds and maintain a frame rate of at least 30 frames per second (FPS) during typical interactions. Latency in rendering AR models should not exceed 100 milliseconds to ensure smooth and responsive user experience, even when manipulating complex urban layouts.

5.3.2 Compatibility. The app will be developed for Android devices and must support a wide range of devices with varying hardware configurations, including mid-range smartphones with AR capabilities.

5.3.3 Usability. The application must have an intuitive user interface that simplifies navigation and interaction. The user experience should prioritize ease of use, with clearly labelled controls and a straightforward interface design.

5.3.4 Scalability. While the initial version will focus on basic urban layouts, the app should be scalable to allow for future expansions, such as integration with real-time traffic data, user-generated content, or the inclusion of additional cities and regions.

5.3.5 Maintainability. The app's codebase should be structured to facilitate easy maintenance and updates, enabling developers to add new features or fix bugs efficiently. It should also be compatible with future versions of Unity and ARCore.

5.3.6 Security. While the app does not handle sensitive personal data, it must adhere to data protection best practices and remain compliant with relevant regulations such as South Africa's POPI Act. All user interactions should be secure, and any data transmitted between the app and external servers (e.g. for data layer integration) must be encrypted using standard protocols (e.g. HTTPS or TLS).

5.4 UML Class Diagram

The UML Class Diagram in Figure B (Appendix B) for SmartCity AR outlines the system's key components and their interactions. It shows how the ARSceneManager handles infrastructure layers and info overlays, the ScenarioManager loads planning scenarios, and the UIManager processes user input. The InfrastructureLayer manages visibility, while the InfoOverlay displays relevant data, ensuring smooth information flow within the AR system.

6 DESIGN

6.1 Data/Class Design

6.1.1 Class diagram. The class diagram in Figure 2 outlines the core structure of SmartCity AR. At the centre is the ARAppController, which manages user interaction, data flow, and visualization. The User class tracks the user's position and interaction history, while the UIController handles toggles for switching between planning modes and infrastructure layers. The EnvironmentZone class organizes various precinct areas and contains ARObject instances representing elements like roads, green spaces, and buildings. This structure supports real-time interaction and comparison of sustainable and unsustainable planning in the Foreshore Freeway Precinct.

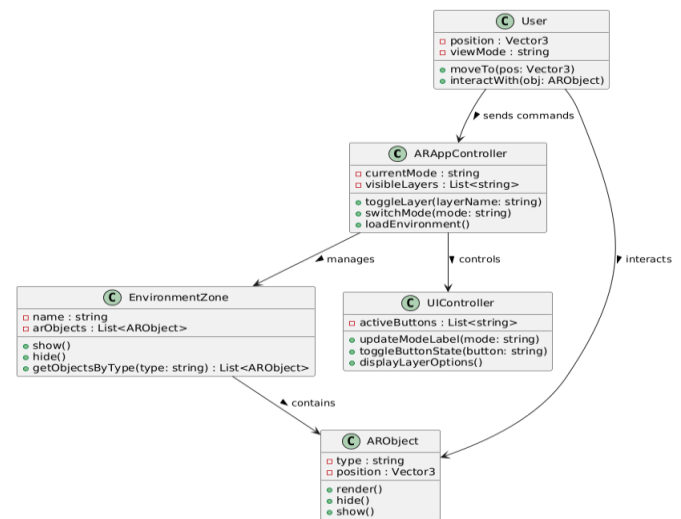


Figure 2: Class diagram showing the main entities and their relationships in SmartCity AR, including user interactions, UI control, and visualization of environment zones and AR objects.

6.1.2 Data flow. The data flow shows how the user interacts with SmartCity AR. By switching between infrastructure layers and modes, the user can enter commands through the UI controller. The AR application controller processes these inputs by retrieving relevant object and environment data from the environment zone components. The AR application gives the user real-time feedback and visualization by updating the visual output in the AR scene based on this data and user actions. Within the program,

this flow guarantees a smooth interplay between user input, data processing, and visual presentation.

6.1.3 State management. The user is initially in an idle state, from which they can interact with interface elements or navigate the precinct. Roads and mixed-use zones are examples of infrastructure layers that the user can toggle, entering a layer toggled state and then going back to idle. Users can also engage with objects, completing interactions before going back to idle.

6.1.4 Design patterns. This helps create a clean and maintainable structure. The Observer Pattern [9] can manage UI updates when users toggle layers like roads or walkways. The State Pattern [10] handles user interactions and view changes between sustainable and unsustainable modes. The Singleton Pattern [11] ensures components like the data manager or session controller remain consistent across the app. These patterns support modularity and make future updates easier to implement [12].

6.2 Architectural Design

6.2.1 The high-level architecture diagram in Figure F (Appendix F) outlines the major system modules of SmartCity AR and their responsibilities. The User Interface handles user controls, including toggling infrastructure layers and switching between sustainable and unsustainable planning views. The Scene Manager is responsible for rendering the 3D city layout and responding to user interactions by showing or hiding relevant models. The Data Layer stores information about city infrastructure, planning data, and sustainability metrics. The State Manager tracks the user's position and interaction state to ensure responsive behaviour. The AR Engine manages camera input, surface detection, and the placement of virtual models in the real world. The Analytics Tracker logs user behaviour and interactions for future evaluation. Together, these modules create a modular and scalable architecture suitable for interactive urban planning in AR.

6.2.2 System boundaries. The system boundaries are defined by its core functionality allowing users to interactively explore urban planning scenarios in augmented reality. The app interfaces with external components such as AR SDKs (e.g. AR Foundation/ARCore) [13] for surface detection and scene placement, as well as a lightweight database to manage infrastructure data and user preferences. Although initially designed for mobile AR, the system can be built with future hardware integration in mind, such as head-mounted AR devices or VR support.

6.2.3 Performance. The system must respond to user interactions within 100 milliseconds and maintain frame rates of at least 30 frames per second (FPS) on supported Android devices. Efficient 3D model rendering and lightweight asset use are critical to meet these performance constraints on mobile devices [14].

6.2.4 Scalability and modularity. These are embedded into the architecture by separating concerns into distinct modules (e.g. UI, Scene Manager, Data Layer), allowing for easier maintenance and potential expansion. Additional layers like traffic data or environmental impact assessments can be integrated without disrupting the core system [14].

6.2.5 Architectural styles. Alternative architectural styles were considered, including monolithic and microkernel designs [15]. A modular, layered architecture was chosen for its balance between simplicity and flexibility. It allows for parallel development of features, easier debugging, and adaptability to evolving requirements and technologies.

6.3 Interface Design

The user interface is centred around intuitive, touch-based interactions suitable for mobile AR. Users will interact with the system by tapping the on-screen buttons to toggle between infrastructure layers (e.g. roads, public transport, mixed-use zones) and switch between sustainable and unsustainable planning views. The UI will include clearly labelled buttons and icons overlaid on the AR scene, allowing for straightforward navigation without obstructing the immersive urban model [12].

Users can quickly load the scene, select infrastructure types, and compare planning scenarios with a few taps. Feedback mechanisms include visual signals (such as highlighted elements or transitions when toggling layers) and subtle audio cues to confirm actions, ensuring users feel engaged and in control.

The interface is designed with user experience goals in mind, emphasizing comfort through minimal on-screen clutter, immersion by allowing interaction directly within the 3D environment, and accessibility by using large, easy-to-tap elements and clear visual language. These considerations support a smooth, enjoyable experience for a wide range of users exploring urban planning designs [14].

6.4 Component Level Design

SmartCity AR consists of modular components that manage interaction and visualization. Core components include the Interaction Manager (handles button inputs and toggling actions), the UI Handler (renders on-screen elements), and the Scene Renderer (displays 3D models based on user input). Supporting modules like the Data Manager store configuration data for scenarios, and the Feedback Engine provides audio or visual cues during interaction.

The component-level diagram in Figure 3 shows how these modules communicate. Inputs are captured by the UI Handler, processed by the Interaction Manager, and relayed to the Scene Renderer and Feedback Engine. Unity is used for prototyping key interactions without full implementation. Each component has defined inputs and outputs to guide development.

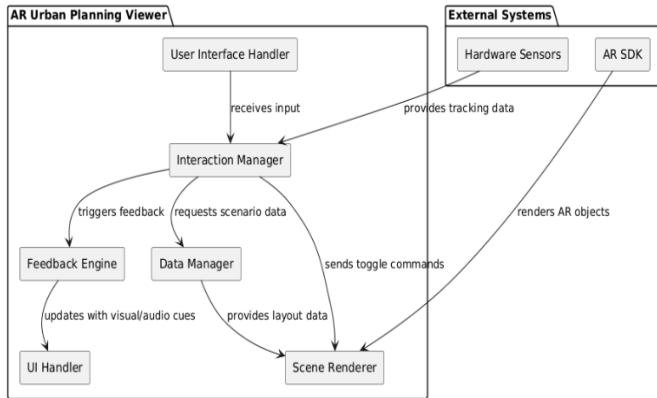


Figure 3: Component-level overview showing communication between core modules of the AR Urban Planning Viewer.

Technical risks include performance limits when rendering multiple 3D elements and varying hardware compatibility. These will be addressed during iterative development and integration with AR SDKs.

7 PROTOTYPE

This project uses a throwaway prototype approach [12] to quickly explore design ideas, interface layouts, and user interactions before full implementation. Unity 2022.3.60f1 will be used to develop the prototype, which will help test and refine core concepts. The prototype is not intended to be reused in the final application but will support early design validation and feedback. For the final system, Unity 6.1 will be used to implement the complete application. This approach allows for rapid iteration and reduces risk by refining design choices early in the development cycle.

7.1 Hardware

No specialized hardware is required for the prototype. It will be tested on a laptop with internet access to run Unity. Later versions may be visually tested on smartphones to simulate AR interface scaling and layout.

7.2 Software

The prototype will be built using Unity 2022.3.60f1. Unity offers a visual development environment with tools like prefab-based design and component-driven workflows, enabling rapid prototyping of interactive features with minimal coding required. It also supports integration with AR Foundation, allowing for cross-platform AR development, including surface detection, model placement, and user interaction.

The prototype will be compiled into a downloadable APK file for Android devices. This will allow users to install and interact with the app directly on their smartphones, enabling hands-on testing of key features such as toggling infrastructure layers (roads, transport, mixed-use zones), switching between sustainable and unsustainable planning views, and navigating the interface. While

the APK will not include full backend logic at this stage, the goal is to demonstrate interactive AR functionality and user flow in a functional, low-code environment.

7.3 Prototype Screenshot

To provide a visual reference for the current state of development, **Figure 4** shows a screenshot of the AR prototype interface built in Unity 2022.3.60f1. This early-stage prototype demonstrates key features such as infrastructure layer toggling and layout navigation. It supports the throwaway prototyping process by validating interface ideas and spatial design choices before full implementation.



Figure 4: Screenshot of the prototype AR interface developed in Unity.

8 CONCLUSION

The development of an Augmented Reality Urban Planning Viewer presents a transformative approach to city planning, addressing the limitations of traditional static tools. By enabling interactive visualization of sustainable and unsustainable urban planning scenarios, this tool enhances decision-making for urban planners, architects, and civil engineers. Focusing on Cape Town's Foreshore Freeway Precinct, the SmartCity AR facilitates the exploration of transport hubs, pedestrian infrastructure, and mixed-use developments in an immersive environment. At this stage, the project is in the prototyping phase, with core features such as infrastructure layer toggling and AR scene placement being developed using Unity and AR Foundation. A downloadable APK will soon be available for testing on Android devices.

The next steps include:

- Integrating user feedback from initial prototype testing.
- Expanding functionality to include basic data layers and UI refinements.
- Preparing for usability evaluation and further stakeholder engagement.

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APPENDICES

APPENDIX A Project Timeline

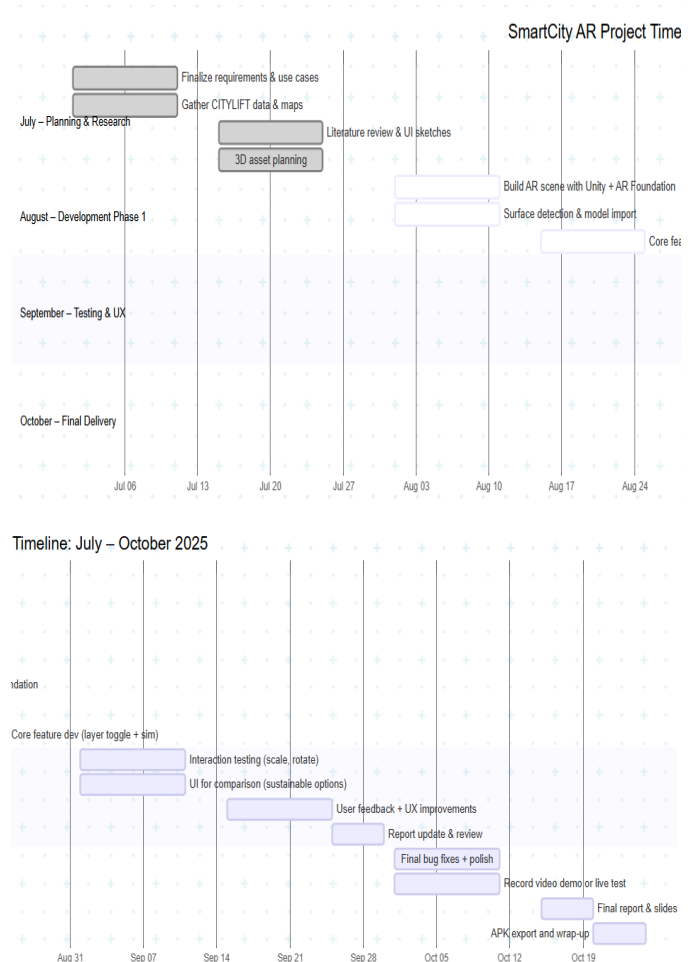


Figure A: The projected timeline of the project.

APPENDIX B

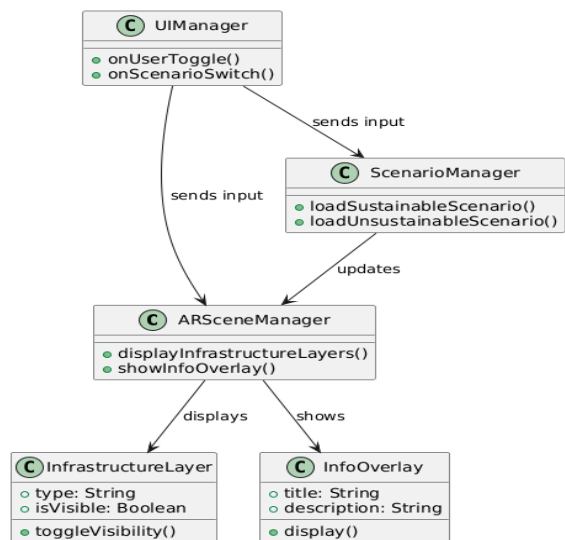


Figure B: UML Class Diagram represents the core classes, their attributes, methods, and relationships in the system.

APPENDIX C
Foreshore Freeway Precinct



Figure C: Illustrates the unfinished highway in the foreshore freeway area.

APPENDIX D
Proposed Solutions



Figure D: CITYLIFT (a proposed master plan for Cape Town) proposes a dense, vibrant, green new city district at the harbour's edge.



Figure E: Illustrates another proposed solution. This proposal is focused on creating affordable housing units and making the city centre accessible and open to everyone. It aims to take the urban wastelands which runs rife in the city centre and turn them into dignified streets.

APPENDIX F

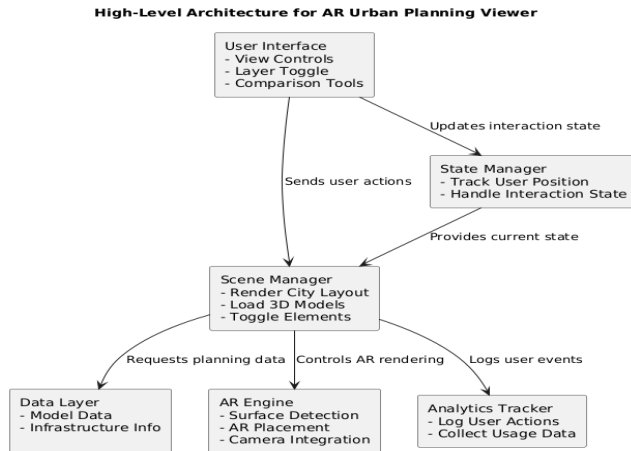


Figure F: Overview of system modules and their responsibilities.

APPENDIX G

GitHub Repository

Link:

<https://github.com/mohamedasad10/SmartCity-AR>

APPENDIX H

Google Sites Website

Link:

<https://sites.google.com/myuwc.ac.za/smartcityar-masad-bandarkar/home>